University of Asia Pacific
Department of Computer Science and Engineering
Final Examination, Fall 2025
Program: B.Sc. in CSE
3rd year 2nd Semester

Course Title: Microprocessors and Microcontrollers Course Code: CSE 315
Time: 3:00 hours Credit Hour: 3.00 Full Marks: 50

There are **Five*** Questions. Answer all of them. Part marks are shown in the margins.

QUESTION 1 [10 MARKS]			
a.	A company is developing two electronic systems: System A: An automatic traffic signal control system that continuously reads sensor inputs and controls traffic lights in real time with minimal external hardware. System B: A computer-based billing and inventory management system for a supermarket. The system requires a keyboard, monitor, large memory, and external storage devices. I. **Determine** which system (A or B) should use a microprocessor and which should use a microcontroller. Give reasons for your answer. II. **Describe** the basic block diagrams of the two processing units used in System A and System B. Give one example of each and compare them.	[2+3]	CLO1/PLOa
b.	Find the time delay for the following delay subroutine if the system has an 8051 with a frequency of 11.0592 MHz. DELAY: MOV R5,#100 — Machine Cycle: 1 BACK: MOV R2,#200 — Machine Cycle: 1 AGAIN: MOV R3,#250 — Machine Cycle: 1	[5]	CLO2/PLOb

	HERE: NOP 1 NOP 1 DJNZ R3,HERE 2 DJNZ R2,AGAIN 2 DJNZ R5,BACK 2 RET 2		
QUESTION 2 [10 MARKS]			
a.	A simple calculator program accepts two single-digit numbers from the keyboard through console input. The system then computes the summation of the entered numbers and displays the result on the screen. Construct an assembly language program to perform the operation.	[10]	CLO3/PLOb
	OR		
b.	Construct an assembly language program to multiply the contents of register AX by 26 using shift instructions and necessary arithmetic operations. Show the calculation performed in each step of the program. Similar as below Construct an assembly language program to read any letter from keyboard and check the case. Moreover, the program will convert case and display with appropriate message.	[10]	CLO3/PLOb
QUESTION 3 [10 MARKS] You are given the following problem statement: Input three integer values x, y, and z (You may choose any suitable values of x and y as input, maintaining the given conditions) from the user, where $0 \leq x,y \leq 4$, $x > y$, and z can take the following values: If $z = 0$, perform addition and display $x + y$ If $z = 1$, perform subtraction and display $x - y$			

If $z = 2$, perform $x \wedge y$ and display the result
If $z = 3$, perform NOT x and display the result
Assume all flags are initially cleared (0) before execution.
 	
List the 8051 Register Banks and their RAM Addresses. State the contents of the RAM locations of 8051 after the following program:
`MOV R0,#99H`
`MOV R1,#85H`
`MOV R2,#3FH`
`MOV R7,#63H`
`MOV R5,#12H`			

a.	**Construct** an equivalent assembly language program to implement the above logic.	[6]	CLO3/PLOb
b.	For each instruction in your assembly program, **determine** which flags (CF, ZF, SF, OF, PF, AF) are affected and show their status (0 or 1) after execution. What is PSW register of 8051? Which bits of the PSW are responsible for selection of the register banks? On power-up, what is the location of the first stack? In the 8051, which register bank conflicts with the stack? In the 8051, what is the size of the stack pointer (SP) register? On power-up, which of the register banks is used?	[4]	CLO2/PLOb
QUESTION 4 [10 MARKS]			
a.	A smart environmental monitoring system receives 50 bytes of sensor data from an SD card module and stores them in memory for further processing. The data transfer takes place sequentially in the forward direction. Construct an assembly language program using string instructions to perform the data transfer operation and briefly illustrate the transfer process. Assume: DS = 3000H, ES = 8000H, DI = 1234H, SI = 3456H, DX = 3567H (port address), , DF = 0 Write an assembly language program that sends 50 bytes of data from consecutive locations of a section of memory (data segment) to another section of memory (extra segment). Assume DS=3000H, ES=8000H, DI=1234H, SI=3456H, DF=0	[5]	CLO3/PLOb

b.	A group of students is developing a smart medicine dispenser for elderly patients as their mini-project. The system should automatically rotate a compartment at specific times of the day so that the correct medicine slot comes in front of the user. The students also want the device to show short messages such as medicine time, warnings, and system status while using as few microcontroller pins as possible. The team is evaluating different types of motors and display modules for the project. I. Select appropriate motor and display modules for this project with justification, including possible problems if a stepper motor and a 16X2 LCD module were used instead. II. Suggest two proper interfacing precautions the students should follow to prevent unstable motor operation and display communication errors.	[3+ 2]	CLO4/PLOc
QUESTION 5 [10 MARKS] The University of Asia Pacific (UAP) plans to develop a smart campus classroom management system to reduce power consumption and improve classroom monitoring. Students and teachers will enter the university using RFID-based ID cards so that entry and exit times can be recorded automatically. Inside the classrooms, the system should monitor human presence, room temperature, humidity, and lighting conditions. If no human presence is detected for a certain period, the system should automatically switch OFF lights, fans, and air conditioners to save energy. The system should also detect smoke or gas leakage in laboratories and activate an alarm during emergency situations. In emergency conditions, warning messages should automatically be sent to students and university authorities through wireless communication.			
a.	Identify the suitable sensors, modules, and actuators for designing this system and justify their use.	[3]	CLO4/PLOc
b.	Illustrate the block diagram of the proposed system.	[3]	CLO4/PLOc
c.	Construct the pseudocode for the working procedure of the system.	[4]	CLO4/PLOc

You can pick questions from below list

	Briefly explain the functions of the following pairs of pins and signals of 8086 – BHE and A0 – HOLD and HLDA – ALE – INTR and INTA – DEN and DT/R – CLK & RESET – MN/MX
	Draw a simplified diagram of a system (personal computer) based on 8086 running in the maximum mode, briefly describe.
	Draw a simplified diagram of a system (personal computer) based on 8086 running in the minimum mode, briefly describe.
	Explain, why buffers are needed in designing system based on 8086? What do you understand by multiplexed bus system, state its merits and demerits? What do you understand by fully demultiplexed fully buffered system?
	Write a program for 8051 to (a) clear accumulator, then (b) add 3 to the accumulator 10 times.
	Write a program for 8051 to (a) load the accumulator with the value 55H, and (b) complement the ACC 700 times.